



Calculation header

Identifier

New calculation (1)

Medium selection and state

Medium

Water

State

Liquid

Operating data

☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	70.0	✗ 0.0	✗ 0.0	°C
p1	10.0	✗ 0.0	✗ 0.0	bar(a)
○ p2	7.0			bar(a)
● Δp	3.0	✗ 0.0	✗ 0.0	bar
Cv	330.07	✗	✗	GPM(US)
○ qm	489,080.0			kg/h
● qv	500.0			m³/h
LpAe	✗ 88.9			dB(A)
	✗ <i>Incipient cavitation</i>			

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ρ1	978.16	✗ 0.0	✗ 0.0	kg/m³
ρ2	978.16			kg/m³
● η1	0.404	✗ 0.0	✗ 0.0	mPa s
○ v1	0.41302			mm²/s
pv1	0.31201	✗	✗	bar(a)
tv1	179.88	373.95	373.95	°C
cF1	1,556.5			m/s

Pipe downstream of valve

Size class downstream of valve

NPS2

2"

Schedule downstream

SCH2

STD



Calculation Sheet

Valve configuration

Valve type	<i>Straight globe valve</i>
Trim type	<i>Cage trim</i>
Flow direction	<i>FTO</i>
Valve performance class	<i>Heavy duty valve</i>
Protection	<i>Hardened seat/plug</i>
Low-noise design	<i>n.a.</i>

Valve data

Basic characteristic		<i>Equal percentage</i>	
☐ Lift stopper			
Selected valve size	NPS	<i>2"</i>	
Pressure class	class	<i>class 600</i>	
Maximum operating pressure	p,max	<i>105.0</i>	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	FL ²	<i>0.644</i>	-
Valve modifier (Cv/Cv100 = 0.75)	Kc	<i>0.516</i>	-
Cavitation modifier (Cv/Cv100 = 0.75)	xFz	<i>0.217</i>	-
Valve style modifier (Cv/Cv100 = 1)	Fd	<i>0.41</i>	-

Load-dependent values

		Maximum flow	Mean flow	Minimum flow	
u2	!	<i>68.525</i>			m/s
LpAe	!	<i>88.9</i>			dB(A)
Ri	!	<i>2.5</i>			-

Noise calculation

Calculation standard (noise, liquid)	<i>IEC 60534-8-4 (2015)</i>
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Noise prediction data

Low-noise design		<i>n.a.</i>	
Sound velocity	cF1	<i>1,556.5</i>	m/s
Pressure level	LA1	<i>166.0</i>	dB
Pressure level	LA2	<i>2.44</i>	dB
Sound pressure level	Lpi	<i>168.0</i>	dB





Calculation Sheet

Noise prediction data (continued)

Transmission loss	TL	-60.1	dB
Sound power level (A-weighted)	LwAe	106.0	dB(A)
Sound pressure level of valve (A-weighted)	LpAe	88.9	dB(A)

Comments:

Reliability index - Ri

Sound pressure level: 22 %; High velocity: 78 %
Recommended solution: Heavy duty valve, Low-noise

Legend

- Errors
- Alarm

